

Pranav Kumar Soma

AI / RESEARCH ENGINEER • AGENTIC SYSTEMS • LLM EVALUATION

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San Francisco Bay Area, CA • Available January 2027

EDUCATION

M.S. Computer Science, University of California San Diego GPA 4.00/4.00 • Jacobs School of Engineering • completed in 1 year	Jan 2026 – Dec 2026 La Jolla, CA
B.S. Computer Science, University of California San Diego GPA 3.96/4.00 • Cum Laude, CSE Department Honors, Provost Honors every quarter • completed in 2 years	Sep 2023 – Dec 2025 La Jolla, CA

EXPERIENCE

AI Research Fellow & Agentic Systems Lead, Laboratory for Emerging Intelligence, UCSD Teradata-funded • Dr. Ramamohan Paturi, Matthew Clegg, Dr. Elizabeth Lyons	Nov 2024 – Present La Jolla, CA
- Lead 6 engineers building an agentic workbench that executes enterprise workflows (100+ pilot users): 60+ MCP servers covering all 600 UCSD APIs, retrieved or auto-generated by pipeline, deployed on GCP Cloud Run - Designed a planner/executor architecture where the planner emits workflow steps and dependencies and an executor harness runs them deterministically, recovering gracefully when a step fails - Built a dual skill and workflow refinement loop driven by production usage, splitting each skill into a description (optimized for retrieval) and a body (optimized for execution) against objectives on latency, token spend, skill-selection accuracy, and redundant human-in-the-loop queries - Architected trace-based context capture with an LLM context manager summarizing user preferences and parameters into hierarchical memory; evaluated with SME, sampling-driven, and heuristic evals - Fine-tuned tutoring LLMs (LoRA, SFT, RLHF) and shipped the AI Tutor to 3 universities across 10 courses; a controlled study over 600+ students measured 64% fewer TA overtime hours, 87% higher engagement. Built its 5-model panel-of-judges eval stack: a reviewer LLM gating every student-facing response and a meta-tutor rewriting the tutor prompt from long-term signal - Research LLM reasoning (mechanistic, multilevel, and interfield) on BioVeritas, scoring logical rigor and analytical soundness in biology papers	
Graduate Researcher, Qualcomm Institute / Calit2, HXI Lab AILA, the AI Living Assistant • Dr. Eliah Aronoff-Spencer	Nov 2024 – Dec 2025 San Diego, CA
- Built multimodal pipelines converting sensor, audio, and video into structured health-database queries, and a conversational interface mapping patient interaction to symptom recognition, risk assessment, and threat analysis - Grounded diagnosis in a MongoDB vector store of UMLS codes (GraphRAG over the medical ontology) and fine-tuned local Qwen models on the San Diego Supercomputer Center against a physician-authored golden eval set. First-author lab poster	
Honors & M.S. Thesis Researcher, UCSD CSE, Computational Redistricting Advisors: Dr. Deepak Kumar, Dr. Imani Munyaka, Dr. Scott Desposato	Sep 2024 – Present La Jolla, CA
Trained XGBoost regressors with multilevel graph partitioning (pooled MAE 0.042, 9 states, held-out cross-state generalization) and implemented a Sequential Monte Carlo sampler (spanning-tree balanced cuts, N=128) over Census 2020 data at 158K to 5.5M blocks/state	
Software Engineer Intern, ServiceNow AI Voice Agents • Technical Skills Award , top intern of 800+	Jun 2025 – Sep 2025 San Diego, CA
Re-architected the NLU pipeline across 8 intents and 48+ slots onto higher-capacity base models: 45% lower agent-response latency, validated live with an enterprise customer. Return offer, returned Summer 2026 to ship AI Voice Agent Analytics at 10K+ calls/day	

PUBLICATIONS & PRESENTATIONS

ORCA: Orchestration and Recognition for Composition and Arrangement. Paper, San Diego Tech Conference 2025 (**Best Project**, 1 of 30 from 200+); poster, SoCal AI Responsibility Summit 2025.

In preparation: computational redistricting (ACM CHI); comparative study of CNNs, VLMs, and MedGemma fine-tuning for on-device lesion detection.

Directed as founder of CSES Innovate: THIA (SDTC 2025) and VITON (SAIRS 2025), presented by the research group.

SELECTED PROJECTS

- AgentSpace** • interview-derived skill and personality models served as hireable agents over two-way **GraphRAG** with **MCP + A2A** • **AIDE** • intent-based IDE using hierarchical vector embeddings and ontologies for regenerative coding and debugging

LEADERSHIP & HONORS

Co-Founder & President, Real World AI Network • **7,000+ student** federation, **50+ faculty** • **President**, CSE Society @ UCSD • **Founder**, CSES Innovate research group (30+ researchers, UCSD/UCLA/UCSB).

Honors: ServiceNow Technical Skills Award • Cum Laude & CSE Department Honors • YC Startup School 2026 • National Merit Finalist • USACO Silver.

TECHNICAL SKILLS

ML & AI: PyTorch, Transformers, scikit-learn, XGBoost, OpenCV; LLM fine-tuning (LoRA, SFT, RLHF), RAG and GraphRAG, vector search, LLM agents, MCP, A2A, LLM-as-judge evals, panel-of-judges, prompt optimization, multimodal models, CNNs, LSTMs, RL, NLP/NLU, computer vision, Sequential Monte Carlo; Qwen, Ollama, MedGemma

Languages & Data: Python, C++, C, Java, TypeScript, JavaScript, SQL, Bash; NumPy, Pandas, GeoPandas, pymetis, Parquet; MongoDB (vector), PostgreSQL; multi-node GPU training, parallel computing, SD Supercomputer Center

Infrastructure: FastAPI, Flask, Node.js, Express, React; Docker, Kubernetes, Jenkins CI/CD, GCP (Cloud Run), AWS (Lex, Lambda), Git, Linux, ROS; distributed and real-time systems, system design